Our final submission utilizes a **Horizon-Decaying UCB1** policy. We chose this algorithm because it perfectly balances the need to distinguish between ads with extremely close, low-yield Click-Through Rates (CTRs) while aggressively eliminating late-game regret by leveraging the known 5000-impression budget.

This final policy was the result of multiple iterative tests, moving from probabilistic to deterministic models to handle the specific quirks of this environment.

The Iteration Journey

1. Standard Thompson Sampling

We started with standard Thompson Sampling using Beta distributions.

- "It is the classic, gold-standard approach for stationary Bernoulli bandits."
- "Result: Scored ~376.6."
- "The Issue: The CTRs of the top three ads were extremely close. Because Thompson Sampling relies on random sampling, the overlapping probability curves caused it to repeatedly draw 'lucky' samples from the 2nd or 3rd best ad, wasting impressions long after it should have committed to the winner."

2. Standard UCB1

To fix the "sloppy exploration" of Thompson Sampling, we switched to UCB1 (Upper Confidence Bound).

- "We wanted a deterministic algorithm that calculates a strict mathematical ceiling for how good an ad could plausibly be, avoiding random lucky draws."
- "Result: Scored ~336.6 (A drop in performance)."
- "The Issue: The overall environment CTR was very low (around 7-8%). In an environment where rewards are this sparse, UCB1's mathematical uncertainty bonus proved too aggressive. It spent thousands of impressions second-guessing itself because bad ads naturally go long stretches without clicks."

3. Horizon-Aware Models & Decaying Epsilon-Greedy

Realizing we needed to control the exploration budget manually, we experimented with algorithms that respected the 5000-impression horizon.

- "We tried Thompson Sampling with informative priors (assuming a ~7.6% starting CTR instead of 50%) and a hard cutoff that forced pure exploitation for the last 40% of the game."
- "We also tested a decaying epsilon-greedy approach where exploration decayed proportionally to 1/(sqrt. t)."
- "Result: Scores stabilized around 385-388."

- "The Issue: Hard cutoffs are a gamble. If the true best ad suffers a brief string of bad luck exactly when the cutoff hits, the algorithm permanently locks onto the wrong ad for the rest of the run."

The Final Algorithm: Horizon-Decaying UCB1

To crack the top tier (scoring ~391.6 and hitting Rank #4), we combined the deterministic bounds of UCB with a dynamically shrinking exploration parameter.

Instead of a sudden cutoff where exploration stops, our policy smoothly reduces the intensity of exploration with every single click. We modified the standard UCB formula by introducing a decaying multiplier c, which scales down as we approach the 5000-impression horizon:

$$c = 0.55 \times \left(1.0 - \frac{t}{\text{horizon}}\right)$$

$$UCB = \hat{\mu}_i + c\sqrt{\frac{\ln(t)}{N_i}}$$

Why this works best:

- "By step 1, the exploration multiplier is high (c is approx 0.55), allowing the algorithm to rigorously test all ads and quickly drop the bottom performers."
- "By step 4000, c is tiny, meaning the algorithm relies almost entirely on the proven historical averages (□) to distinguish between the top two close contenders."
- "By step 5000, c = 0, forcing pure exploitation exactly when the value of learning new information is zero."

This method natively adapts our risk tolerance to the remaining runway, yielding the highest consistency across varying hidden seeds without relying on rigid cutoffs.